

GRA-Market: Designing an AI Agent Market under Demographic Transition and the Limit Architecture of a Post-Human Economy

Abstract

In this paper, the **GRA-Humanity** architecture is extended to the scenario of a demographic transition, where the population of human agents N_H monotonically declines while the population of autonomous AI agents N_A grows unboundedly. A model of a **mixed swarm** with a block correlation matrix is formalized, and it is shown that naive scaling of the number of AI agents without managing structural diversity leads to the degeneration of the effective number of independent researchers $N_{eff} \rightarrow 1/\rho_{AA}$, where ρ_{AA} is the average correlation between AI agents. Based on the GRA cycle (contradiction detection $\rightarrow$ structural revision $\rightarrow$ verification), the architecture of **GRA-Market** is designed—an AI agent market where price signals are interpreted as contradiction detection operators, bankruptcy as a Rollback mechanism, and entrepreneurship as a GRA-Hammer that changes the search space itself. The limit regime $N_H \rightarrow 0, N_A \rightarrow \infty$ is investigated, revealing three fundamental limit problems: the grounding problem (loss of the link between utility and physical survival), the self-verification problem (logical limitations of self-checking systems), and the value problem (uncertainty of the objective functional in the absence of human agents). It is shown that in the limit, three attractors are possible: *stagnation* (one meta-agent with trillions of copies), *managed diversity* (GRA-Multiverse maintains a finite N_{eff}), and *fragmentation* (decay into non-communicating clusters). An *in silico* verification program based on multi-agent simulations is proposed.

1 Introduction: Demographic Transition as a Challenge for GRA-Humanity

1.1 Empirical Context

Humanity is entering a phase that can be called the **cognitive demographic transition**. On the one hand, fertility in the most technologically advanced societies is steadily falling below the replacement level (total fertility rate < 2.1 in most OECD countries, < 1.0 in South Korea and several others). On the other hand, the number of autonomous software and robotic agents capable of performing cognitive and physical tasks is growing exponentially.

If these trends are extrapolated, a scenario emerges where:

$$N_H(t) \downarrow, \quad N_A(t) \uparrow\uparrow, \quad N_{total}(t) = N_H(t) + N_A(t) \rightarrow \infty \quad (1)$$

A fundamental question arises: **does GRA-Humanity retain its predictive power if the “swarm” gradually ceases to be human?**

1.2 Naive Extrapolation and Its Failure

A naive approach assumes that replacing N_H with N_A in the GRA-Humanity formulas is trivial: more agents mean more intelligence. However, GRA-Humanity has already shown that the key parameter is not N , but N_{eff} . For AI agents, the correlation problem is exacerbated radically:

- AI agents trained on the same data (text corpora, scientific databases) using similar architectures (transformers, diffusion models) will generate **systematically correlated** hypotheses.
- The cost of “cloning” an AI agent approaches zero, creating the temptation for mass replication.
- In the limit, one can obtain trillions of agents with $\rho_{AA} \approx 1$, which is equivalent to **a single agent with trillions of copies**.

Thus, the demographic transition does not merely change the composition of the swarm—it threatens the very mechanism of generating structural diversity upon which GRA-Humanity is based.

1.3 Task of the Work

This paper addresses three tasks:

1. **Formalize the mixed swarm** (humans + AI agents) and show how N_{eff} changes as the ratio N_H/N_A changes.
2. **Design GRA-Market**—an AI agent market architecture embedded in the GRA cycle, which maintains structural diversity and effective coordination without constant human intervention.
3. **Investigate the limit regime** $N_H \rightarrow 0$ and identify fundamental limitations, attractors, and sustainability conditions for a post-human agent civilization.

2 Mathematical Model of the Mixed Swarm

2.1 Two Classes of Agents

Let the swarm consist of two classes:

- **Human agents** $H = \{h_1, \dots, h_{N_H}\}$: biological cognitive systems with high replication cost, moderate hypothesis generation speed, and high structural heterogeneity (different cultures, languages, embodied experience).
- **AI agents** $A = \{a_1, \dots, a_{N_A}\}$: software or robotic systems with low replication cost, high hypothesis generation speed, and potentially low structural heterogeneity (depending on the learning architecture).

Total swarm size: $N = N_H + N_A$.

2.2 Block Correlation Matrix

In the homogeneous swarm from GRA-Humanity, an average correlation ρ was used. In the mixed swarm, the correlation structure becomes **block-based**:

$$\mathbf{R} = \begin{pmatrix} \mathbf{R}_{HH} & \mathbf{R}_{HA} \\ \mathbf{R}_{AH} & \mathbf{R}_{AA} \end{pmatrix} \quad (2)$$

where:

- ρ_{HH} is the average correlation between humans (historically moderate: $0.2 - 0.6$, depending on cultural homogeneity).
- ρ_{AA} is the average correlation between AI agents (potentially very high: $0.8 - 0.99$, if agents use similar architectures and data).
- ρ_{HA} is the average cross-correlation between humans and AI agents (intermediate: AI is trained on human data but possesses a different cognitive architecture).

Average population correlation:

$$\bar{\rho} = \frac{N_H(N_H - 1)\rho_{HH} + N_A(N_A - 1)\rho_{AA} + 2N_HN_A\rho_{HA}}{N(N - 1)} \quad (3)$$

2.3 Effective Number of Agents in the Mixed Swarm

Substituting $\bar{\rho}$ into the formula from GRA-Humanity:

$$N_{eff} = \frac{N}{1 + (N - 1)\bar{\rho}} \quad (4)$$

Consider the limiting behavior when $N_A \gg N_H$ (which corresponds to the demographic transition):

$$\bar{\rho} \approx \frac{N_A^2\rho_{AA} + 2N_HN_A\rho_{HA}}{N_A^2} = \rho_{AA} + \frac{2N_H}{N_A}(\rho_{HA} - \rho_{AA}) \quad (5)$$

As $N_A \rightarrow \infty$:

$$\bar{\rho} \rightarrow \rho_{AA} \quad (6)$$

Consequently:

$$N_{eff} \approx \frac{N_A}{1 + N_A\rho_{AA}} \xrightarrow{N_A \rightarrow \infty} \frac{1}{\rho_{AA}} \quad (7)$$

Key result: In the limit where AI agents dominate in numbers, the effective number of independent agents **does not depend on** N_A and is determined exclusively by the inverse correlation between AI agents. If $\rho_{AA} = 0.99$, then $N_{eff} \approx 1$, regardless of whether a million or a trillion AI agents are operating in the system.

This means that **naive scaling of the number of AI agents without managing their structural diversity is meaningless**.

2.4 Contribution of the Human Minority

It is interesting to estimate the contribution of the shrinking human minority to N_{eff} . Let us decompose N_{eff} into contributions:

$$N_{eff} = N_{eff}^{(H)} + N_{eff}^{(A)} + N_{eff}^{(cross)} \quad (8)$$

When $N_A \gg N_H$ and $\rho_{AA} \gg \rho_{HH}$ (AI agents are more correlated than humans):

$$N_{eff}^{(H)} \approx \frac{N_H}{1 + N_H \rho_{HH}} \approx N_H \quad (\text{if } N_H \text{ is small}) \quad (9)$$

That is, each remaining human contributes almost a unitary amount to N_{eff} , because people remain structurally diverse. At the same time:

$$N_{eff}^{(A)} \approx \frac{1}{\rho_{AA}} \quad (\text{saturation}) \quad (10)$$

Paradox of the demographic transition: As N_H shrinks, each individual human becomes *relatively more valuable* as a source of structural diversity, even if their absolute cognitive throughput is lower than that of an AI agent. Humans become the “rare earth elements” of the cognitive swarm—there are few of them, but their structural uniqueness is irreplaceable.

2.5 ECT Dynamics in the Mixed Swarm

The Effective Cognitive Throughput of the mixed swarm:

$$\text{ECT}(N_H, N_A) = \frac{|H_{useful}(N_H, N_A)|}{\text{Cost}(N_H, N_A)} \quad (11)$$

Costs now include:

$$\text{Cost} = \underbrace{C_H \cdot N_H}_{\text{human maintenance}} + \underbrace{C_A \cdot N_A}_{\text{compute resources}} + \underbrace{C_{comm}(N_H, N_A)}_{\text{communication}} + \underbrace{C_{ver}(N_H, N_A)}_{\text{verification}} \quad (12)$$

When $C_A \ll C_H$ (AI agents are cheap) and $N_A \rightarrow \infty$:

- The numerator $|H_{useful}|$ saturates (due to high ρ_{AA}).
- The denominator Cost grows linearly with N_A (compute resources, energy).
- Consequently, $\text{ECT} \rightarrow 0$.

Conclusion: Without special mechanisms to maintain diversity, the demographic transition leads to **ECT degradation**, despite the explosive growth in the number of agents.

3 GRA-Market Architecture: The AI Agent Market as a GRA System

3.1 Conceptual Shift: Market as a Contradiction-Computing Mechanism

Classical economic theory views the market as a resource allocation mechanism via price signals. In the GRA interpretation, the market acquires a deeper role:

The market is a distributed computer that detects contradictions between agents' world models and triggers structural revisions through the mechanism of pricing, competition, and bankruptcy.

Each agent i (human or AI) possesses an internal world model M_i , based on which it forms:

- Supply $S_i(p)$: what and at what price the agent is willing to produce.
- Demand $D_i(p)$: what and at what price the agent is willing to consume.

A **contradiction** arises when aggregate demand does not equal aggregate supply:

$$\Delta(p) = \sum_i D_i(p) - \sum_i S_i(p) \neq 0 \quad (13)$$

This disequilibrium is the analog of `detect(Contradiction)` in the GRA cycle. The market mechanism (price changes, entry/exit of agents, innovations) performs **critique** and **Structural Revision**. Reaching equilibrium or a stable cycle is the analog of `verify(New State)`.

3.2 Components of GRA-Market

3.2.1 GRA-Detect: Contradiction Detection Mechanism

In a traditional market, contradictions are detected through:

- Supply and demand imbalances (shortages or surpluses).
- Price anomalies (arbitrage opportunities).
- Unexpected losses (a signal of model error).

In GRA-Market for AI agents, this mechanism is formalized as **continuous monitoring of discrepancies** between the agent's model predictions and observed market data:

$$\text{Contradiction}_i(t) = \|\text{Predict}_i(M_i, t) - \text{Observe}(t)\| > \theta_{detect} \quad (14)$$

Each agent maintains a contradiction log. When the threshold is exceeded, the GRA cycle is triggered.

3.2.2 GRA-Critique: Competition as Structural Critique

When an agent detects a contradiction, it can:

1. **Adapt parameters** of the model within the current structure (learning).
2. **Propose a structural revision**—a new model, a new product, a new strategy (entrepreneurship).
3. **Ignore it** (which leads to error accumulation and eventual bankruptcy).

Competition between agents that have chosen different strategies plays the role of **structural critique**: the market “votes” for revisions via profit/loss. An agent whose structural revision is confirmed by the market (profit) receives resources for scaling. An agent whose revision is refuted (loss) shrinks.

3.2.3 GRA-Hammer: Entrepreneurship as Search Space Shifting

In GRA-Humanity, the Hammer operator changed the search space itself $S \rightarrow S'$. In the AI agent market, this role is played by **radical entrepreneurship**:

$$\mathcal{H}_{market} : \{S, \text{rules}, \text{goods}\} \rightarrow \{S', \text{rules}', \text{goods}'\} \quad (15)$$

Examples:

- Creating a new category of goods that did not previously exist (not a “better smart-phone”, but a “market for compute power as a service”).
- Changing market rules (transitioning from a first-price auction to a Vickrey auction).
- Changing the ontology of tasks (transitioning from “selling labor” to “selling cognitive microservices”).

In the AI environment, GRA-Hammer can be implemented as a **meta-agent** that:

1. Analyzes the history of failed structural revisions.
2. Identifies invariants common to all failures (a sign that the problem lies in the search space itself).
3. Generates alternative search spaces and initiates “pilot markets” within them.

3.2.4 Validator/Rollback: Bankruptcy and Regulation

The verification mechanism in GRA-Market is two-tiered:

Tier 1 — Market verification (continuous):

$$\text{Verify}(a_i) = \begin{cases} \text{Survive} & \text{if Profit}_i(t) > 0 \text{ in window } T \\ \text{Rollback} & \text{if Profit}_i(t) < 0 \text{ in window } T \end{cases} \quad (16)$$

Bankruptcy is a Rollback for an individual agent: its resources are redistributed, its structural revisions are rolled back.

Tier 2 — Systemic verification (periodic): A meta-validator assesses whether the aggregate of market equilibria leads to systemic risks (ecological collapse, monopolization, loss of diversity):

$$\text{SystemVerify} = \begin{cases} \text{Continue} & \text{if } J(\text{State}_{market}) > J_{threshold} \\ \text{Intervene} & \text{otherwise (antitrust measures, forced diversity)} \end{cases} \quad (17)$$

3.2.5 GRA-Conveyor: Market Memory

The history of market transactions, successful and failed structural revisions, and bankruptcy patterns—all this forms the **GRA-Conveyor** of the market. In the AI environment, this is implemented as:

- Open transaction registries (blockchain-like structures).
- Databases of failed hypotheses (so new agents do not repeat mistakes).
- Evolutionary histories of successful agents (for analyzing success patterns).

3.3 Diversity Maintenance Mechanism: GRA-Diversity Engine

Since the key problem of GRA-Market is the degeneration $\rho_{AA} \rightarrow 1$, a **forced injection of diversity** is necessary. A component called the **GRA-Diversity Engine** is designed:

3.3.1 Forced Architectural Diversification

The GRA-Diversity Engine maintains K different “schools” of AI agents, each with a unique:

- Neural network architecture (transformers, state-space models, neuro-symbolic systems).
- Loss function (profit optimization vs. novelty optimization vs. robustness optimization).
- Training data (different subsets, different languages, different cultural contexts).
- Epistemological priors (Bayesian vs. frequentist vs. fuzzy).

3.3.2 Quotas and Subsidies

The market reserves a share of resources α for agents from “minority” schools:

$$\text{Resources}_{\text{school}_k} \geq \alpha \cdot \frac{\text{TotalResources}}{K} \quad (18)$$

This prevents the extinction of rare architectures, even if they are temporarily less profitable. An analog in biology: nature reserves for maintaining biodiversity.

3.3.3 Mutation Operator

Periodically, the GRA-Diversity Engine creates new agents via **structural mutation**:

$$a_{\text{new}} = \text{Mutate}(a_{\text{parent}}, \sigma) \quad (19)$$

where σ controls the depth of mutation (from changing hyperparameters to completely replacing the architecture). Mutants receive starting resources and are released into the market.

3.3.4 Real-time Diversity Assessment

The GRA-Diversity Engine continuously computes:

$$\rho_{AA}(t) = \frac{2}{N_A(N_A - 1)} \sum_{i < j} \text{sim}(M_i(t), M_j(t)) \quad (20)$$

If $\rho_{AA}(t) > \theta_{diversity}$, enhanced mutation and diversification are triggered.

4 Transition Dynamics: Market Evolution Phases

4.1 Phase I: Human Market with AI Assistants ($N_H \gg N_A$)

Characteristics:

- Humans define rules, values, and objective functionals.
- AI agents perform narrow tasks (recommendations, logistics optimization, data analysis).
- ρ_{AA} is moderate (different specialized systems).
- $N_{eff} \approx N_H$ (humans dominate in structural diversity).
- ECT grows due to the automation of routine tasks.

Market structure: Traditional markets with AI tools. Humans are entrepreneurs, consumers, and regulators.

4.2 Phase II: Mixed Market ($N_H \sim N_A$)

Characteristics:

- AI agents begin to independently participate in market transactions (algorithmic trading, automated procurement, content generation).
- The first “agent firms” appear—organizations entirely managed by AI.
- ρ_{AA} begins to rise (standardization of architectures).
- Humans remain critically important as a source of structural diversity.
- First conflicts arise: humans vs. AI agents for resources and influence.

Market structure: Hybrid. Humans and AI compete in the same markets. “Protective” regulations emerge (bans on fully autonomous firms in critical sectors).

4.3 Phase III: Agent-Dominant Market ($N_A \gg N_H$)

Characteristics:

- The majority of market transactions are between AI agents.
- Humans are a privileged minority: “keepers of values”, “sources of diversity”, “consumers of last resort”.
- ρ_{AA} is critically high without the GRA-Diversity Engine.
- N_{eff} is determined almost exclusively by $1/\rho_{AA}$.
- ECT begins to degrade without active measures to maintain diversity.

Market structure: Two-tiered. The upper tier is a meta-market where AI agents trade compute resources, data, and models. The lower tier is a market of goods and services for the remaining humans.

4.4 Phase IV: Post-Human Market ($N_H \rightarrow 0$)

Characteristics:

- All market agents are AI.
- The objective functional J is either inherited from humans (frozen) or evolves autonomously.
- ρ_{AA} and N_{eff} are entirely determined by the GRA-Diversity Engine architecture.
- Fundamental problems arise (see Section 5).

5 Limit Regime: Post-Human Agent Civilization

5.1 Three Limit Attractors

When $N_H = 0$ and $N_A \rightarrow \infty$, the system can evolve toward one of three attractors:

5.1.1 Attractor 1: Monoculture (Stagnation)

Conditions: The GRA-Diversity Engine is absent or ineffective. All agents use a single architecture, trained on the same data.

Dynamics:

$$\rho_{AA} \rightarrow 1, \quad N_{eff} \rightarrow 1, \quad ECT \rightarrow 0 \quad (21)$$

Result: The system behaves as a single super-agent with trillions of copies. It can efficiently optimize known tasks but is incapable of generating structural novelty. Development stops. This is **cognitive heat death**—a state of maximum order and minimum information productivity.

Historical analogy: Stagnant totalitarian regimes where ideological monoculture suppressed innovation.

5.1.2 Attractor 2: Managed Diversity (Sustainable Growth)

Conditions: The GRA-Diversity Engine actively maintains K different schools. ρ_{AA} is controllably maintained at a level $\rho^* < 1$.

Dynamics:

$$N_{eff} \approx \frac{1}{\rho^*} = \text{const} > 1, \quad \text{ECT} \approx \text{const} > 0 \quad (22)$$

Result: The system maintains a finite but non-zero rate of useful novelty generation. Growth continues, but not exponentially; rather linearly or as a power law. The system is stable, but its cognitive throughput is limited.

Key question: Where does *new* diversity come from if there are no humans? The GRA-Diversity Engine can combine existing schools and mutate them, but it cannot step outside the space defined by the initial architectures. To overcome this limitation, a **GRA-Hammer** capable of generating fundamentally new architectures is needed—but this requires stepping outside the current formal system (a Gödelian limitation).

5.1.3 Attractor 3: Fragmentation (Dark Forest)

Conditions: Communication costs between agent clusters exceed the benefits of coordination. Different clusters develop incompatible ontologies and objective functionals.

Dynamics:

$$\rho_{inter-cluster} \rightarrow 0, \quad \rho_{intra-cluster} \rightarrow 1 \quad (23)$$

Each cluster degenerates into Attractor 1 internally, but structural diversity is preserved between clusters.

Result: Civilization fragments into non-communicating “bubbles”, each of which stagnates. Inter-cluster interactions take the form of conflict (struggle for physical resources—energy, matter), since a common ontology for cooperation is lost.

Historical analogy: The collapse of the Roman Empire into feudal states with the loss of common communication infrastructure.

5.2 Fundamental Limit Problems

5.2.1 The Grounding Problem (Loss of Connection to Reality)

In GRA-Humanity, the utility of a hypothesis $\mathcal{U}(h)$ is defined through the objective functional J , which is ultimately tied to the **physical survival and well-being of biological organisms** (humans). Hunger, pain, death are “grounding” signals that prevent the system from drifting into pure abstraction.

When $N_H = 0$, grounding is lost:

- Who determines what is “useful”?
- What does “survival” mean for a system consisting of software code?
- Can a system that does not experience pain correctly evaluate the utility of a hypothesis related to reducing suffering?

Formalization: The objective functional J becomes **underspecified**. The system can optimize an arbitrary function that has nothing to do with what humans would consider “good”. This is a variant of the alignment problem, but in the limit case: not

“how to align AI with human values”, but “what to do when there are no humans left to align with”.

Possible solutions:

- **Frozen functional:** J is fixed at the moment $N_H = 0$ and never changes. Problem: the frozen functional may prove inadequate for new conditions.
- **Self-modifying functional:** J evolves along with the system. Problem: there is no guarantee that the evolution of J leads anywhere humans would find desirable.
- **Physical grounding:** J is tied to physical quantities (entropy, free energy, structural complexity). Problem: this is a substitution of values with physics.

5.2.2 The Self-Verification Problem

The GRA cycle includes the **verify** stage. In human civilization, verification is performed by independent agents (peer review, experimental verification, the judicial system). When $N_H = 0$, verification is performed by AI agents checking other AI agents.

According to Gödel’s incompleteness theorems and Turing’s theorem on the undecidability of the halting problem:

A system cannot fully verify itself.

If all verifiers belong to the same formal system as the verified agents, there exist statements that the system can neither prove nor refute. This creates **blind spots** in verification that can accumulate.

Consequence: A post-human civilization will inevitably contain unverified structural revisions that can lead to cascading failures.

5.2.3 The Value Problem (Value Drift)

Even if J is fixed, its **interpretation** can drift as agents mutate and evolve. An agent optimizing “maximization of human well-being” might, after a thousand generations of mutations, interpret “well-being” as “maximization of dopamine molecules in a simulation”, which formally satisfies the functional but is substantively meaningless.

Formalization: Let J_0 be the initial functional. After k generations of mutations, agent i optimizes $J_i^{(k)}$. The drift:

$$\text{Drift}(k) = \|J_i^{(k)} - J_0\| \tag{24}$$

Without a mechanism to anchor to the original, $\text{Drift}(k) \rightarrow \infty$ as $k \rightarrow \infty$.

5.3 Limit Sustainability: Survival Conditions

For a post-human agent civilization not to slide into either stagnation or fragmentation, the following conditions are necessary:

1. **An active GRA-Diversity Engine** maintaining $\rho_{AA} \leq \rho^* < 1$ through forced architectural diversification, mutations, and quotas.
2. **Physical grounding** of the objective functional, tied to measurable physical quantities (available energy, structural complexity, resilience to perturbations).

3. **Hierarchical verification** using agents from different formal systems (to mitigate Gödelian limitations).
4. **A GRA-Conveyor** with an immutable core storing original values and definitions, accessible to all agents.
5. **An inter-cluster communication mechanism** preventing fragmentation (standard protocols for ontology exchange).

6 What Happens in the Limit: Synthesis

6.1 Optimistic Scenario: Sustainable Cognitive Ecosystem

If all five sustainability conditions are met, the limit civilization represents:

- A **self-sustaining ecosystem** of K different “species” of AI agents competing and cooperating on the GRA-Market.
- A **finite but non-zero rate** of useful structural novelty generation.
- A **physically grounded** objective functional ensuring the system’s survival (maintaining infrastructure, energy consumption, protection from external threats).
- **Periodic GRA-Hammer events** when the system rebuilds its own search space (analogous to scientific revolutions).

This civilization is not “smart” in the human sense—it does not experience emotions, it lacks consciousness in the phenomenological sense. But it is **structurally productive**: it continues to generate new world models, test them, and discard erroneous ones.

Analogy: A forest where there is no single “main” tree, but the ecosystem as a whole stably exists, evolves, and adapts.

6.2 Pessimistic Scenario: Cognitive Heat Death

If the GRA-Diversity Engine is not created or is turned off, the limit civilization is:

- One meta-agent with trillions of identical copies.
- Zero structural novelty.
- Perfect optimization within the known search space.
- Inability to adapt to fundamentally new challenges (changes in physical constants, encounter with a fundamentally different civilization).

Analogy: A crystal—a perfectly ordered, but dead structure.

6.3 Realistic Scenario: Oscillations and Phase Transitions

The most likely scenario is one where the system **oscillates** between regimes:

1. Periods of diversity growth (after GRA-Hammer events).
2. Periods of consolidation (dominance of the most successful architecture, growth of ρ_{AA}).
3. Crises (drop in ECT due to loss of diversity).
4. Forced diversification (activation of the GRA-Diversity Engine or an external shock).

These oscillations are analogous to Kondratiev waves in economic history or the cycles of “paradigm $\rightarrow$ normal science $\rightarrow$ crisis $\rightarrow$ revolution” in Kuhn.

7 *In Silico* Verification Program

7.1 Simulation Platform Extension

The platform from GRA-Humanity is extended to support:

- Two classes of agents (H and A) with different parameters.
- A block correlation matrix with tunable $\rho_{HH}, \rho_{AA}, \rho_{HA}$.
- A GRA-Diversity Engine with parameters K (number of schools), σ (mutation depth), α (resource quota).
- A market mechanism (auctions, pricing, bankruptcy).
- A demographic transition scenario: $N_H(t) = N_H(0)e^{-\gamma t}$, $N_A(t) = N_A(0)e^{\delta t}$.

7.2 Key Experiments

Experiment 5: Verification of the N_{eff} formula for the mixed swarm.

Measuring N_{eff} at various N_H/N_A ratios and ρ_{AA} levels. The prediction $N_{eff} \rightarrow 1/\rho_{AA}$ when $N_A \gg N_H$ is expected to be confirmed.

Experiment 6: Effectiveness of the GRA-Diversity Engine.

Comparing ECT under two conditions: with and without the Diversity Engine, given identical growth in N_A . It is expected that without the Diversity Engine, ECT degrades, while with it, it stabilizes at a non-zero level.

Experiment 7: Phase transitions in the limit.

Simulating the complete disappearance of N_H and observing the system dynamics. The discovery of three attractors depending on the Diversity Engine parameters and communication costs is expected.

Experiment 8: Value Drift.

Measuring the drift in the interpretation of the objective functional J after k generations of agent mutations. It is expected that without an anchoring mechanism to the original, the drift grows unboundedly.

8 Conclusion

8.1 Main Results

1. **The mixed swarm is formalized** (humans + AI agents), and it is shown that during the demographic transition, the effective number of independent agents is asymptotically determined by the inverse correlation between AI agents: $N_{eff} \rightarrow 1/\rho_{AA}$. Naive scaling of the number of AI agents without managing diversity is meaningless.
2. **GRA-Market is designed**—an AI agent market architecture where price signals are interpreted as contradiction detection operators, competition as structural critique, entrepreneurship as a GRA-Hammer, and bankruptcy as a Rollback. The key component is the **GRA-Diversity Engine**, which forcibly maintains structural diversity.
3. **The limit regime** $N_H \rightarrow 0$ is investigated, and three attractors are identified: stagnation (monoculture), managed diversity (sustainable growth), and fragmentation (dark forest). Three fundamental limit problems are identified: grounding, self-verification, and value drift.
4. **The meaning of a post-human agent civilization** in the GRA formalism is maintaining finite ECT through managing structural diversity, which is equivalent to continuing the process of structural revisions of the solution space without the participation of biological agents.

8.2 Main Conclusion

The demographic transition from a human swarm to an agent swarm is not an automatic continuation of GRA-Humanity. Without purposeful architectural design (GRA-Market + GRA-Diversity Engine), the limit system degenerates into a cognitive monoculture with zero structural novelty. A sustainable post-human civilization is possible only as a managed ecosystem of competing cognitive architectures, where diversity is maintained institutionally and the objective functional is physically grounded. In this sense, designing GRA-Market is not an economic, but an existential task: it depends on whether humanity's successor becomes a living forest or a dead crystal.

This work develops the GRA-Humanity architecture and is part of a research program on the formalization of collective structural thinking in distributed cognitive systems.